

NOVEMBER - DECEMBER 2025 EXAMINATION  
I Yr. B.TECH. (BM, CE, CS, EE & EC Branches only)  
PH10009: APPLIED PHYSICS

Time : 3 hours

Max Marks : 70

TOTAL NUMBER OF SECTIONS IN THIS PAPER - 05.

NOTE: Missing data if any, may be assumed appropriately.

Given:  $\hbar = 1.054 \times 10^{-34}$  Js,  $k_B = 1.38 \times 10^{-23}$  J/K,  $m_e = 9.11 \times 10^{-31}$  kg and  $c = 3 \times 10^8$  m/s.

| Q. No. | QUESTION                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Marks | CO | BL  | PI |
|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|----|-----|----|
| 1      | (a) With neat sketch discuss various light - matter interaction processes.                                                                                                                                                                                                                                                                                                                                                                                                   | 2     | 1  | 2   | 1  |
|        | (b) Match the following:<br>(i) Population Inversion ( $\alpha$ ) Helps perform round trip of light.<br>(ii) A Pump ( $\beta$ ) $\text{Al}_2\text{O}_3:\text{Cr}^{3+}$ crystal in Ruby laser.<br>(iii) Resonator ( $\gamma$ ) Electric discharge in He-Ne laser<br>(iv) An Active medium ( $\delta$ ) $N_1 \leq N_2$                                                                                                                                                         | 2     | 1  | 1,3 | 1  |
|        | (c) Obtain an expression for the ratio of Einstein's A and B Coefficients.                                                                                                                                                                                                                                                                                                                                                                                                   | 3     | 1  | 4   | 1  |
|        | (d) (i) What are metastable states? What is their significance in Lasing action?<br>(ii) Why two-level lasers does not exists?<br>(iii) Why photons are called Bosons?                                                                                                                                                                                                                                                                                                       | 3+2+2 | 1  | 4   | 2  |
|        | <b>OR</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |       |    |     |    |
|        | (e) With appropriate energy diagram and schematics, explain the working of a solid state laser and a gas laser.                                                                                                                                                                                                                                                                                                                                                              | 3½+3½ | 1  | 1,2 | 3  |
| 2      | (a) Classify optical fibers based on their refractive index profile and mode profile.                                                                                                                                                                                                                                                                                                                                                                                        | 2     | 2  | 1   | 1  |
|        | (b) Draw a neat ray diagram of light guiding in an optical fiber. Derive an expression for maximum incident angle ( $i_{max}$ ) at the input end of the optical fiber for no-loss light guiding.                                                                                                                                                                                                                                                                             | 2     | 2  | 3   | 2  |
|        | (c) Write an essay on the contribution of any two scientists only, from the following (i) NS Kapany (ii) SN Bose (iii) CKN Patel and (iv) ECG Sudharshan.                                                                                                                                                                                                                                                                                                                    | 3     | 2  | 1,6 | 1  |
|        | (d) (i) A ray of light is launched at $7.5^\circ$ at the air-core interface of an optical fiber of $n_{core} = 1.5$ . For TIR find the minimum value $n_{cladding}$ .<br>(ii) The angle of acceptance of an optical fiber is $30^\circ$ when kept in air. Find the acceptance angle when the same fiber is immersed in a liquid of refractive index 1.33 .<br>(iii) The attenuation in optical fiber is 3.6dB/km. What fraction of it; initial intensity remain after 1.5km. | 3+2+2 | 2  | 4,5 | 3  |
|        | <b>OR</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |       |    |     |    |
|        | (e) (i) Draw the refractive index profiles of step-index and graded index optical fibers.<br>(ii) If $n_0$ , $n_{co}$ and $n_{cl}$ , are the refractive Indices of air, core and cladding, show that it is possible to have $NA > 1$ when $n_0 > 1$ .                                                                                                                                                                                                                        | 3+4   | 2  | 1,3 | 3  |
| 3      | (a) Is earth an inertial frame of reference? What about Sun? Explain.                                                                                                                                                                                                                                                                                                                                                                                                        | 2     | 3  | 2   | 1  |
|        | (b) Find the kinetic energy (KE) in classical theory and in relativistic theory for an electron moving at a velocity of $0.99c$ . Compare both values of KE and discuss.                                                                                                                                                                                                                                                                                                     | 2     | 3  | 3,4 | 2  |
|        | (c) How fast would an athlete need to be running for a 100-m race to look 100-ft long? What is unreasonable? Discuss.                                                                                                                                                                                                                                                                                                                                                        | 3     | 3  | 5   | 2  |

|   |           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |       |   |     |     |
|---|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|---|-----|-----|
|   | (d)       | Obtain expressions for the relation between mass & energy and mass & momentum. Also reduce the expressions of Energy and momentum for a massless particle.                                                                                                                                                                                                                                                                                                                              | 3+3+1 | 3 | 1,3 | 3   |
|   | <b>OR</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |       |   |     |     |
|   | (e)       | What are Galilean and Lorentz transformation equation in $x, y, z$ and $t$ . Using Einstein's postulates obtain Lorentz transformations from Galilean transformations.                                                                                                                                                                                                                                                                                                                  | 2+5   | 3 | 2   | 2,3 |
| 4 | (a)       | Estimate the number of photons per second, emitted from a light source of wavelength 580nm and optical power 10W.                                                                                                                                                                                                                                                                                                                                                                       | 2     | 4 | 5   | 2   |
|   | (b)       | An atom in an excited state stores energy temporarily. If the lifetime of this excited state is measured to be 100 ps, what is the minimum uncertainty in the energy of the state?                                                                                                                                                                                                                                                                                                      | 2     | 4 | 6   | 2   |
|   | (c)       | An electron and a photon of wavelength 500nm has same momentum. Calculate the energy of these electron and the photon and compare.                                                                                                                                                                                                                                                                                                                                                      | 3     | 4 | 3,4 | 3   |
|   | (d)       | (i) What is Compton effect?<br>(ii) Obtain an expression for the wavelength of scattered photon in Compton effect.<br>(iii) If incident X-rays have a wavelength of 10.0 pm, the scattered X-rays at a certain angle have a wavelength of 10.5 pm. Find the momentum (magnitude and direction) of the corresponding recoil electrons.                                                                                                                                                   | 1+3+3 | 4 | 1,3 | 1,2 |
|   | <b>OR</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |       |   |     |     |
|   | (e)       | (i) What are the significance of "Normalization Condition"? Explain. If $\psi = Ae^{-i(Et/\hbar + i\alpha x/2)}$ is a normalized wave function, find the value of A.<br>(ii) Using Schrödinger's equation, obtain a normalised wavefunction for a particle in a one dimensional potential well. Also plot the probability amplitude and probability for $n = 1, 2, 3$ .                                                                                                                 | 2+5   | 4 | 2,4 | 1,3 |
| 5 | (a)       | What are unitary operations? Does, any classical Gate has unitary property?                                                                                                                                                                                                                                                                                                                                                                                                             | 2     | 5 | 6   | 1,2 |
|   | (b)       | What do you understand by the word "entanglement"?                                                                                                                                                                                                                                                                                                                                                                                                                                      | 2     | 5 | 2   | 1   |
|   | (c)       | In the following cases find the $\square$ —,<br>(i) $ 0\rangle \rightarrow \square \text{---} H \text{---} \square \text{---}  0\rangle$<br>(ii) $ 1\rangle \rightarrow \square \text{---} Z \text{---}  1\rangle$                                                                                                                                                                                                                                                                      | 3     | 5 | 6   | 2   |
|   | (d)       | Solve the following:<br>$\begin{array}{l}  \psi_1\rangle \text{---} \square \text{---} H \text{---} \bullet \text{---} \square \text{---} X \text{---}  \phi_1\rangle \\  \psi_2\rangle \text{---} \square \text{---} X \text{---} \oplus \text{---} \square \text{---} H \text{---}  \phi_2\rangle \end{array}$<br>for all possible 2-Qubit inputs ( $ \psi_1\psi_2\rangle$ ) and find output ( $ \phi_1\phi_2\rangle$ ) for each case. Report the results in the form of Truth-Table. | 7     | 5 | 5   | 3   |
|   | <b>OR</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |       |   |     |     |
|   | (e)       | Find the output for all possible inputs ( $ \psi_1\psi_2\rangle$ ):<br>$\begin{array}{l}  \psi_1\rangle \text{---} \square \text{---} H \text{---} \text{---} \bullet \text{---}  \phi_1\rangle \\  \psi_2\rangle \text{---} \square \text{---} H \text{---} \text{---} \oplus \text{---}  \phi_2\rangle \end{array}$<br>Explain the output qubits in each case                                                                                                                         | 7     | 5 | 4   | 2,3 |

CO-Course outcome. PI-Performance Indicator. (1.Cognitive, 2. Affective, 3.Psychomotor). BL-Bloom's Taxonomy Levels: 1-Remember, 2-Understand, 3-Apply, 4-Analyze, 5-Evaluate, 6-Create.